

Project ID :

25-26J-407

1. Topic (12 words max)

Inclusive Educational Software System: Emotionally Expressive Reading for Children with Disabilities (Blind, Deaf, Cognitive Impairments)

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Children with disabilities, such as those who are blind, deaf, or have cognitive impairments, face significant barriers in accessing and understanding educational materials. Traditional educational resources are predominantly designed for children without disabilities, leaving children with special needs with limited options for engaging with stories and study content. This disparity limits their ability to fully participate in educational experiences, which are essential for their cognitive, social, and emotional development.

For blind children, text-based content is a primary challenge, as it requires conversion into braille or audio formats to be accessible. Deaf children encounter difficulties with auditory-based content, which often requires sign language translations, captioning, or visual cues to ensure accessibility. Children with cognitive disabilities struggle to process complex or abstract information, which can become overwhelming if content is not simplified and tailored to their learning needs. As a result, these children often experience delays in educational progress, despite the availability of assistive tools.

While advances in assistive tools have improved accessibility, the focus has often been on addressing the needs of only one disability type at a time. Current systems rarely provide integrated solutions that address the overlapping and diverse needs of blind, deaf, and cognitively impaired children simultaneously. Additionally, existing tools often lack the ability to incorporate emotional expressiveness, a crucial element for maintaining engagement, improving retention, and ensuring comprehension.

Many existing **online self-learning platforms** and resources, such as educational apps, interactive e-books, and video tutorials, provide basic accessibility features like captions or text-to-speech, but they still lack emotional engagement and fail to address the needs of all three disability groups in a cohesive manner. Self-learning tools can be enhanced by including **interactive, emotional cues** in content, allowing children to engage with the material at their own pace, whether it's through personalized emotional feedback or simplified text with emotional context.

This research aims to bridge this gap by creating an **emotionally expressive reading system** designed specifically for children with disabilities. The project seeks to foster a more engaging and inclusive self-learning experience, where children can independently explore educational content, while ensuring that it resonates emotionally and cognitively with their unique learning needs. Ultimately, the goal is to ensure equal educational opportunities for all children, regardless of their abilities.

References :

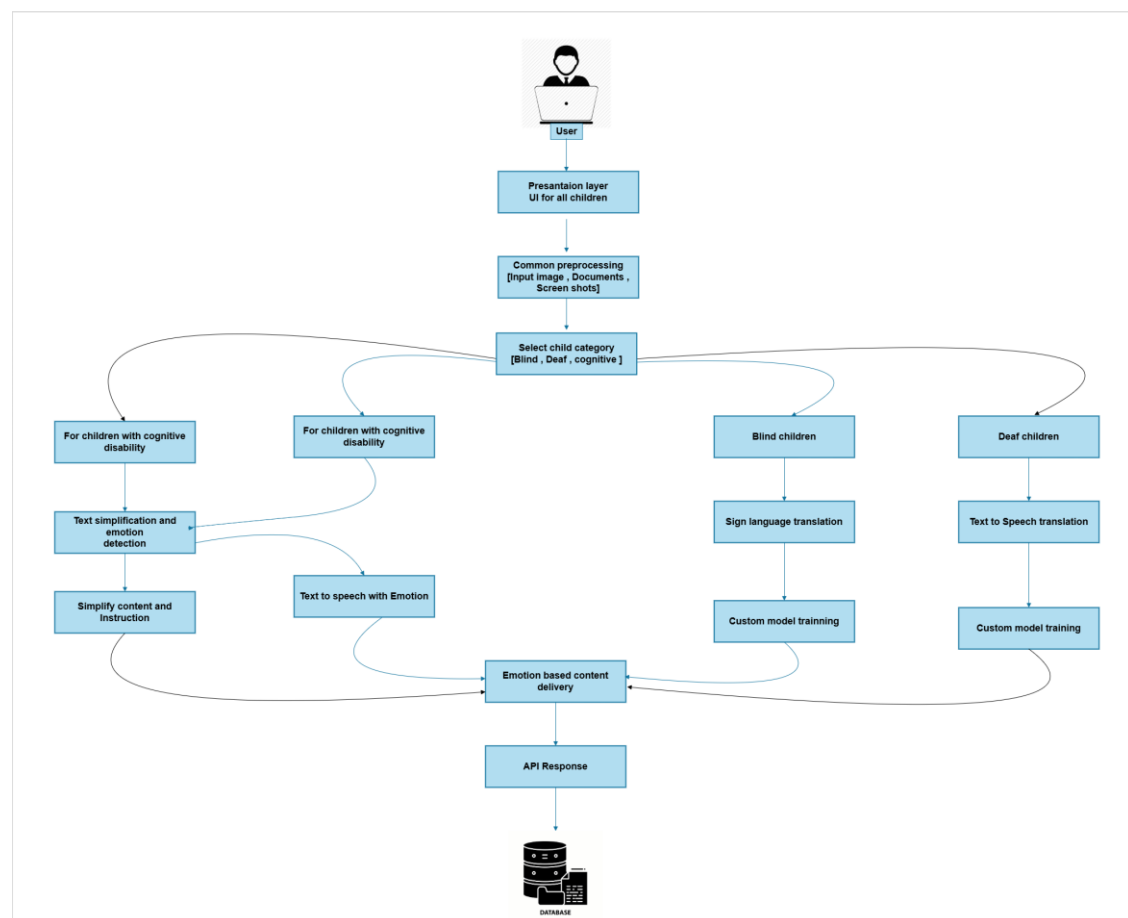
- **M. E. Bausch and T. L. Sapp**, *Understanding Assistive Technology: An Introduction to Students with Disabilities and their Educational Needs*, Pearson, 2008.
- **R. P. Silva**, "Assistive Technology for Children with Learning Disabilities: A Systematic Literature Review," *ResearchGate*, 2023. [Online]. Available: https://www.researchgate.net/publication/368373667_Assistive_Technology_for_Children_with_Learning_Disabilities_A_Systematic_Literature_Review. [Accessed: 22-Jun-2025].
- **UNICEF**, *Country Profile: Disability-Inclusive Education Practices in Sri Lanka*, 2020. [Online]. Available: <https://www.unicef.org/rosa/media/17016/file/Country%20Profile%20-%20Sri%20Lanka.pdf>. [Accessed: 22-Jun-2025].

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is an **emotionally expressive reading system** designed to provide an inclusive learning experience for children with disabilities, such as those who are blind, deaf, or have cognitive impairments. This system focuses on creating an engaging, personalized, and emotionally resonant educational experience, allowing children with varying disabilities to independently access and interact with learning materials.

The system will offer a multi-modal approach, integrating **audio feedback**, **visual cues**, and **simplified text**, all while incorporating emotional expressiveness. For blind children, the system will provide text-to-speech (TTS) with varied emotional tones, helping children connect emotionally with the content. For deaf children, sign language translation will accompany the material, enriched with emotional expressions to enhance understanding. For children with cognitive disabilities, the text will be simplified, and emotional context will be added to make the learning experience more engaging and relatable.

This approach aims to enhance learning engagement and comprehension by providing content that resonates emotionally with each child, ensuring a richer and more personalized learning experience.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

To successfully develop the **emotionally expressive reading system** for children with disabilities, a multidisciplinary approach is required, combining expertise in various fields such as assistive technology, education, natural language processing (NLP), emotion recognition, and multimedia design.

Specialized Expertise:

1. **Assistive Technology:** Expertise in designing and developing **accessible learning tools** for children with disabilities, particularly for **blind, deaf, and cognitively impaired** children. Knowledge of accessible formats like **braille, audio content, and sign language** translation is critical to ensuring inclusivity.
2. **Natural Language Processing (NLP):** Proficiency in **text simplification** and **emotion detection** to make learning materials easier to understand for children with cognitive disabilities. Additionally, expertise in creating **context-aware systems** to adapt content according to emotional cues is essential for maintaining engagement and comprehension.
3. **Text-to-Speech (TTS) & Emotional Expressiveness:** Expertise in implementing **emotionally expressive TTS** systems, capable of adjusting tone, pitch, and speed to align with the emotional context of the learning content. This knowledge is crucial for ensuring that children can emotionally connect with the material.
4. **Sign Language Translation:** A deep understanding of sign language, particularly its multiple dialects, is necessary for developing translation systems. Knowledge of video/avatar animation for sign language output is required to integrate sign language translation into the educational experience.
5. **Educational Design:** Knowledge of special education techniques and learning principles is essential for adapting the content to different cognitive levels and ensuring that the solution is pedagogically sound.

Data Requirements

1. **Educational Content:** Access to a variety of educational materials for different subjects, including stories, texts, and diagrams, to test the system's ability to handle various content types.
2. **Sign Language Data:** A corpus of sign datasets representing various dialects and contexts.
3. **Speech and Emotion Data:** High-quality voice data for training TTS systems with emotional expressiveness and different voice types.
4. **User Feedback:** Data collected from children with disabilities and educators to assess the effectiveness and engagement of the solution.

8. Objectives and Novelty

Main Objective The main objective of this project is to create an emotionally expressive reading system that provides accessible, engaging educational content for children with disabilities, including those who are blind, deaf, or have cognitive impairments. By integrating features like text-to-speech , sign language translation , and simplified text , the system adapts to the child's emotional state to enhance learning and ensure an inclusive, personalized educational experience.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22103222 Vikesh R	Image Acquisition, Preprocessing & OCR Optimization	- Develop image capture/upload UI with voice support. - Preprocess images (noise removal, de-skewing, enhancement). - Fine-tune OCR for different materials. - Design error correction for OCR. - Create preprocessing pipeline for digital/physical content. - Train custom models for handling various learning materials, including storybooks, diagrams, and handwritten content.	- Lack of real-time OCR processing capabilities for varied content types (e.g., handwritten, diagrams). - Need for efficient error correction and optimization in OCR for different learning formats. - Absence of a seamless preprocessing pipeline for both digital and physical materials. - Insufficient training data for diverse educational content to improve OCR model accuracy.
IT22153104 Ilangeeran I	NLP Module: Text Simplification, Summarization,	- Simplify text for children. - Detect emotions in text. - Implement context-aware semantic tagging for multimodal use.	- Lack of emotion-based content adaptation for better engagement. - Insufficient context-aware semantic tagging that facilitates inclusive

	Emotion Detection & Contextualization	- Build APIs to connect text with TTS and sign language. - Train emotion detection models using annotated datasets (e.g., ISEAR dataset) to accurately classify emotional tones in text.	- learning for multimodal content. - Gap in integrating emotion detection for dynamic, interactive learning environments. - Need for more data-driven emotion recognition models using diverse datasets for accuracy.
IT22282590 SENTHEEPAN E	Text-to-Speech (TTS) with Emotional Expressiveness & User Interaction	- Integrate TTS with emotional expressiveness (tone, pitch, pace). - Develop interactive playback controls. - Implement accessibility features for TTS. - Fine-tune TTS models using diverse voice data to produce accurate emotional expressions. - Train voice synthesis models with datasets containing emotional cues to enhance engagement.	- Lack of emotionally dynamic TTS systems adjusting tone, pitch, and pace for better user engagement. - Limited accessibility features in TTS, especially for individuals with disabilities. - Insufficient multilingual support for emotional TTS systems. - Need for more emotional cues in voice synthesis models to improve contextually relevant speech.
It22123954 Brinthapan J	Sign Language Translation & UI Multimodal Integration	- Develop sign language translation models for multiple dialects. - Create sign language translation for educational content. - Design an accessible UI for users. - Implement multimodal interaction (sign language or voice). - Train sign language models with datasets to enhance accuracy and emotional expressiveness in translation..	- Lack of emotionally enhanced sign language translation for richer content delivery. - Limited integration of multimodal interaction combining sign language and voice for an enriched learning experience. - Need for dynamic, user-friendly UI that enhances accessibility. - Gap in training sign language models with emotional context for more accurate translations.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22103222 Vikesh R	Specialization: Assistive Technology & Image Processing role focuses on developing the image capture/upload system with accessibility features for blind users. The tasks, including OCR optimization and image preprocessing , align with their specialization in assistive technologies, ensuring content is accessible to children with visual impairments. The development of error correction mechanisms and adaptive preprocessing pipelines further enhances accessibility across various educational materials, aligning with the goal of providing high-quality content for all learners.
IT22153104 Ilangeeran I	Specialization: Natural Language Processing (NLP) & Cognitive Disabilities Specializes in making text more accessible and emotionally engaging for children with cognitive disabilities. Simplifying and summarizing educational text, as well as detecting emotions, directly aligns with their NLP expertise. This ensures that the content is understandable, emotionally resonant, and adaptable to the learning needs of each child. By implementing context-aware tagging, they bridge the gap between language processing and multimodal learning, fulfilling the specialization's objective of creating an emotionally engaging learning environment.
IT22282590 SENTHEEPAN E	Specialization: Speech Synthesis & Emotional Expressiveness role in developing emotionally expressive text-to-speech (TTS) and user interaction features directly reflects their specialization in speech synthesis. The integration of emotional expressiveness (tone, pitch, pace) into the TTS system ensures a natural, engaging, and personalized learning experience, tailored to each child's needs. Their expertise in accessibility features like voice speed and volume adjustments ensures that children can control their experience based on their comfort and ability.
It22123954 Brinthapan J	Specialization: Sign Language Translation & Multimodal Interaction specializes in the development of sign language translation models and multimodal integration . Their work in creating sign language translations ensures that educational content is accessible to children who are deaf or hard of hearing. The development of emotionally enhanced sign language ensures richer content delivery. Additionally, the integration of multimodal features, such as interactive sign language and voice control, reflects their expertise in delivering a fully accessible and engaging learning experience for all children.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	MS.	Sanjeevi	Chandrasiri	<i>Sanjeevi</i>
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
-----	-------------------------------------	----	--------------------------

- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
-----	-------------------------------------	----	--------------------------

- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
-----	-------------------------------------	----	--------------------------

- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
-----	-------------------------------------	----	--------------------------

- e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.